Why are Diamonds so Expensive?

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Introduction

The international diamond industry relies on the beauty and perceptions of rareness by the consumer. Jewelers market the four Cs, carat, cut, clarity, and color, as having equal value. Through a massive sample of diamonds, we can test whether this marketing assertion conforms to statistical fact. The interpretation of such trends gives an insight into the way retail pricing strategies are established and what consumers are paying.

Objective

The project examines the Diamonds dataset by getting insights into how the features of carat, clarity, and color are affecting the prices of diamonds. This is aimed at determining which of the characteristics influence value the most.

Data Source

Diamonds data set of the ggplot2 library with 53,940 records on diamond features and price.

Variables

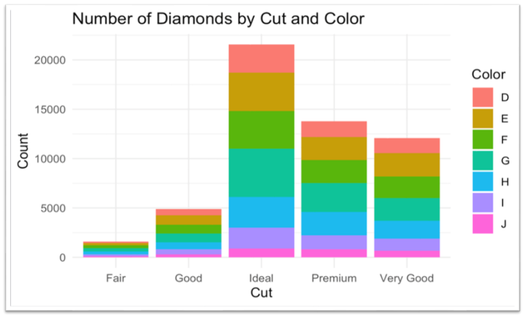
Categorical Variables :

- Cut- determines how rough the diamond is shaped into the finished product. The order of the grades is, "Fair", "Good", "Very Good", "Premium", "Ideal".
- Color- letters D-F represent colorless diamonds while letters G-J represent colored diamonds.
- Clarity- how clear a diamond appears to be. Grades range from "I1" (worst) to "IF" (best).

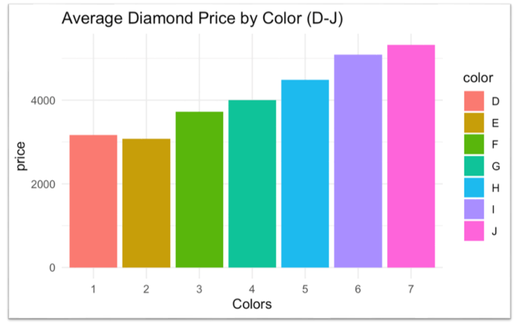
Continuous Variables:

- Carat- refers to the measurement of the diamonds mass, one carat is equal to 0.2 grams.
- Price- Cost in dollars ($) for each diamond
- X, Y, Z, Depth, Table- All various diamond measurements in millimeters (mm).

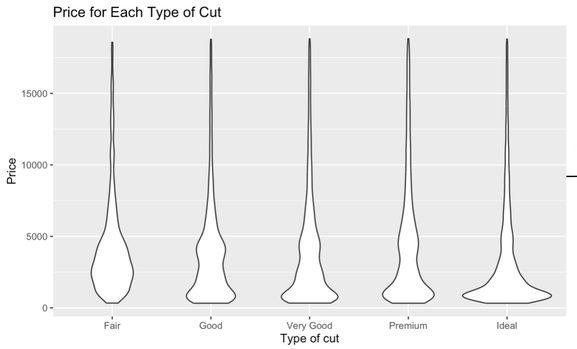
Visualizations and Analyses



When delving into the size of area covered by each grade of color, typically, there are more diamonds in the range of "G" to "J" than "D" to "F". This indicates that there are more colored diamonds than colorless diamonds for almost every single type of cut.
It seems that the most common grade of color is "G" as it typically takes up the most area for each type of cut. This may correlate as to why there are more colored diamonds than colorless diamonds as previously stated above.



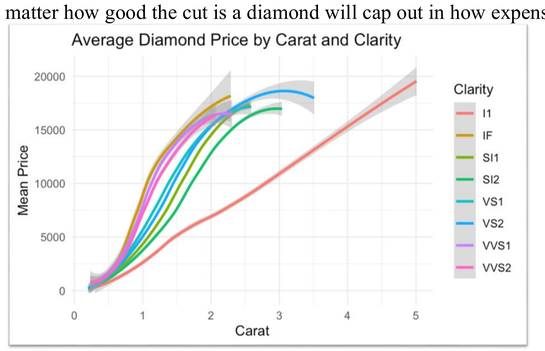
This chart is used to demonstrate the change in the average diamond price with change in color grade. Diamonds that are graded with a closer color grade of J would have higher average prices whereas those with a D or E color would have lower average prices. This can be viewed as a surprising fact because D is the best color grade, yet it can also help to mean that other variables, including carat size or clarity, can have a stronger impact on the price in this dataset.



Sample code

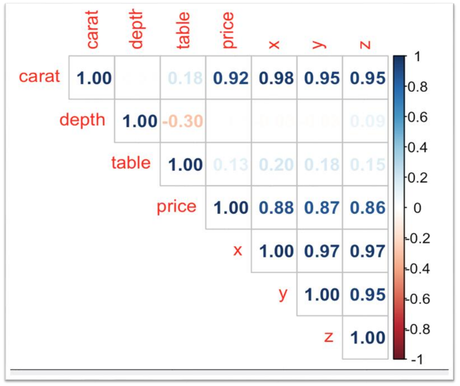
```
Diamonds_factored <- Diamonds %>%
     mutate(factor_cut   =   factor(cut,   levels   =
c("Fair",
"Good",
"Very Good",
"Premium",
"Ideal")))
Diamonds_factored %>%
  ggplot(aes(x = factor_cut, y = price)) +
  geom_violin() +
  labs(x = "Type of cut",
       y = "Price",
       title = "Price for Each Type of Cut")
```

Based off the graph, it seems that there is more diamonds within the 0-5000 range of price for all types of cuts but especially "Fair". This could indicate that there are larger amounts of diamonds in the "Fair" classification that are cheaper than other types of cuts.
It seems like there is relatively even distribution of values between the price of 5000-15000 in all types of cuts. This could indicate that no matter how good the cut is a diamond will cap out in how expensive it can be due to other attributes about said diamond.



In this graph, the price of diamonds increases with carat size. The difference with the I1 clarity diamonds is that they have a significantly lower price than all other levels of clarity, at all carats. This implies that the diamonds that have visible inclusions (I1) are less preferred and the other clarity grades follow the same pattern but at a higher price. In general, the difference between I1 and the rest demonstrates the degree of influence of clarity on diamond value.

Correlation Matrix



The following correlation matrix indicates the relationship among various features of diamonds. There is a positive correlation between the price and carat size (0.92), which implies that bigger diamonds are relatively costly. Both carat and price are also strongly correlated with the dimensions (x, y, z), as bigger diamonds have a more natural tendency to weigh more. Depth and table on the other hand have very weak or negative correlation implying that they do not influence price much compared to size.

Key Findings

In our analysis, we find that carat weight is far the most important factor when determining the price of a diamond. Carat has a strong positive correlation with price which can completely dominate the impact of other attributes. Clarity and color also influence price, but it is significantly less pronounced and more gradual particularly with diamonds of less than one carat. The quality of cut has a moderate impact on price variation, although its influence is less predictable than carat. The correlation matrix also indicates that there is a strong positive relationship between carat and price, but depth and table exhibit weak or insignificant relations. Overall, the data shows that the market values size in the first place, and carat is the best value driver, with all the other features being primarily refinements and not price drivers.